In the prenous class : $V = iR$, $R = \rho \frac{l}{A}$ Ω

$$W_R = \int v\, i\, dt = i_R^2 R \cdot t \quad (J)$$

$$v_L = L \frac{di_L}{dt}; \quad L = \frac{N^2}{\left(\frac{l}{\mu A}\right)} = \frac{N^2}{Rl} \quad (H)$$

$$W_L = \frac{1}{2} L i_L^2 \quad (J)$$

$$v_C = \frac{1}{C}\int i_C dt; \quad C = \frac{\epsilon A}{d} \quad (F)$$

$$W_C = \frac{1}{2} C V^2 \quad (J)$$

Voltage Source, current source
Ideal, Nonideal, dependent sources

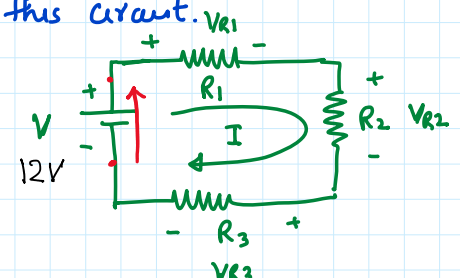
Kirchhoff's Laws: ① Kirchhoff's Voltage Law (KVL)
② Kirchhoff's Current Law (KCL)

Kirchhoff's Voltage Law: The sum of voltage, at any instant is a closed circuit is zero, i.e.,

$$\sum_{j=1}^{k} V_j = 0$$

Mesh analysis

For example consider this circuit.



$R_1 = 1\Omega$, $R_2 = 2\Omega$, $R_3 = 3\Omega$

$-V + V_{R1} + V_{R2} + V_{R3} = 0$

$V = V_{R1} + V_{R2} + V_{R3}$

$VI = V_{R1} I + V_{R2} I + V_{R3} I$

$V\int I\,dt = V_{R1}\int I\,dt + V_{R2}\int I\,dt + V_{R3}\int I\,dt$

$Vq = V_{R1} q + V_{R2} q + V_{R3} q$

$W = W_1 + W_2 + W_3$

$V = R_1 I + R_2 I + R_3 I$

$V = (R_1 + R_2 + R_3) I = R_e I$

$I = \frac{V}{R_e}$; $R_e = (R_1 + R_2 + R_3)$

$R_e = 1 + 2 + 3 = 6\Omega$

$I = \frac{12}{6} = 2\,A$

$W = W_1 + W_2 + W_3$

$R_e = 1+2+3 = 6\Omega$

$I = \frac{12}{6} = 2A$

$V_{R1} = 1 \times 2 = 2V$

$V_{R2} = 2 \times 2 = 4V$

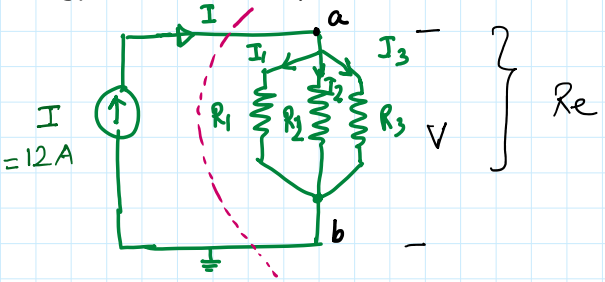
$V_{R3} = 3 \times 2 = 6V$

$V_{R1} + V_{R2} + V_{R3} = 12V = V$

Kirchhoff's Current Law: At any instant, the sum of currents at any junction point in a circuit is zero.

$$\sum_{j=1}^{k} I_j = 0$$

For example consider the circuit,



$R_1 = 1\Omega, R_2 = 2\Omega, R_3 = 3\Omega$

$V_{ab} = ? = V$

$-I + I_1 + I_2 + I_3 = 0$

$I - (I_1 + I_2 + I_3) = 0$

$\frac{V}{R_e} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$

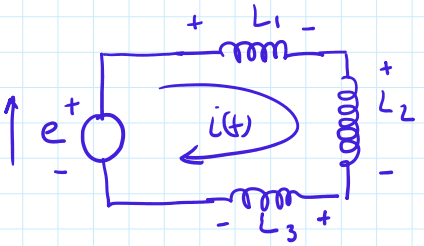
$$\boxed{\frac{1}{R_e} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots}$$

$R_e < (R_1, R_2, R_3)$

$R_e = \frac{R_1 R_2}{R_1 + R_2}$

$R_1 \| R_2$

Series Connected Inductors



KVL: $e = V_{L1} + V_{L2} + V_{L3}$

$= L_1 \frac{di}{dt} + L_2 \frac{di}{dt} + L_3 \frac{di}{dt}$

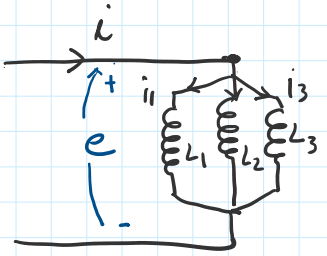
$\therefore \frac{di}{dt} = (L_1 + L_2 + L_3) \frac{di}{dt}$

$$= L_1\frac{di}{dt} + L_2\frac{di}{dt} + L_3\frac{di}{dt}$$

$$L_{eq}\cdot\frac{di}{dt} = (L_1+L_2+L_3)\frac{di}{dt}$$

$$\Rightarrow \boxed{L_{eq} = L_1+L_2+L_3} \checkmark$$





KCL: $i = i_1 + i_2 + i_3$

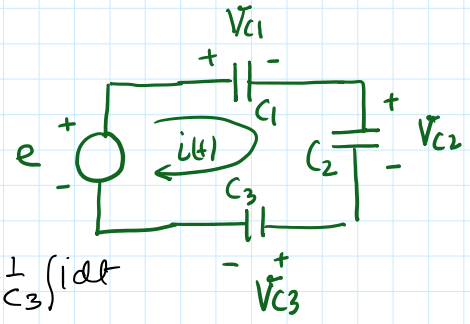
$$\frac{1}{L_{eq}}\int v\,dt = \frac{1}{L_1}\int v\,dt + \frac{1}{L_2}\int v\,dt + \frac{1}{L_3}\int v\,dt$$

$$\Rightarrow \boxed{\frac{1}{L_{eq}} = \frac{1}{L_1}+\frac{1}{L_2}+\frac{1}{L_3}} \checkmark$$

$$v_L = L\frac{di_L}{dt}$$

$$\frac{1}{L}\int v_L\,dt$$

Capacitance Connection

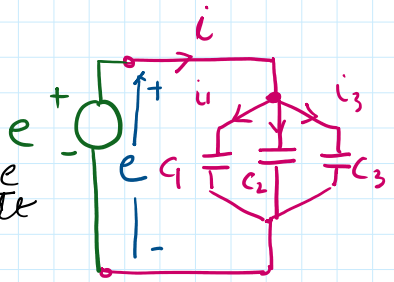


KVL: $e = v_{c1} + v_{c2} + v_{c3}$

$$\frac{1}{C_{eq}}\int i\,dt = \frac{1}{C_1}\int i\,dt + \frac{1}{C_2}\int i\,dt + \frac{1}{C_3}\int i\,dt$$

$$\Rightarrow \boxed{\frac{1}{C_{eq}} = \frac{1}{C_1}+\frac{1}{C_2}+\frac{1}{C_3}}$$ for series Connection.

Shunt Connected Capacitors



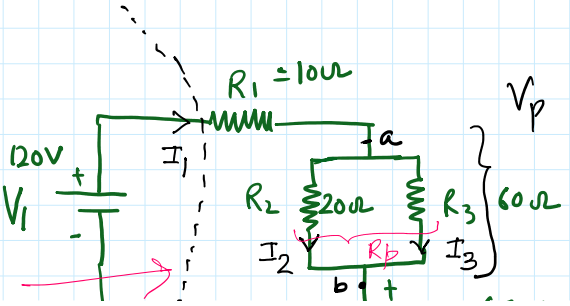
$$i = i_1 + i_2 + i_3$$

$$C_{eq}\frac{de}{dt} = C_1\frac{de}{dt} + C_2\frac{de}{dt} + C_3\frac{de}{dt}$$

$$\boxed{C_{eq} = C_1+C_2+C_3}$$

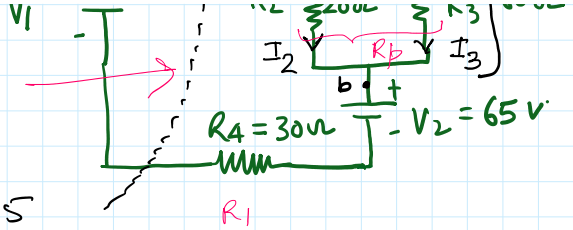
Superposition Theorem

Example 1: Find out the current in R_2 ?



$$R_p = R_2 \| R_3 = \frac{20\times60}{20+60} = \frac{1200}{80} = 15\,\Omega$$

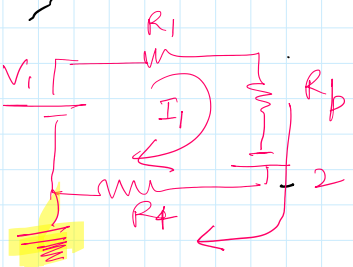
$$R_p = R_2 \| R_3 = \frac{20\times 60}{20+60} = \frac{1200}{80} = 15\,\Omega$$



$$-V_1 + R_1 I_1 + R_p I_1 + V_2 + I_1 R_4 = 0$$

$$I_1 = \frac{V_1 - V_2}{R_1 + R_p + R_4} = \frac{120 - 65}{55}$$

$$\boxed{I_1 = 1\ A}$$



$$V_p = 1 \times 15 = 15\ V$$

$$I_{R_2} = \frac{V_p}{R_2} = \frac{15}{20}\ A = \frac{3}{4}\ A$$

$$I_{R_3} = \frac{V_p}{R_3} = \frac{15}{60}\ A = \frac{1}{4}\ A$$

$$I_1 = \frac{V_1}{R_1 + R_p + R_4} + \frac{(-V_2)}{R_1 + R_p + R_4}$$

$$I_1 = I_{11} + I_{12}$$

Superposition Theorem is stated as follows...

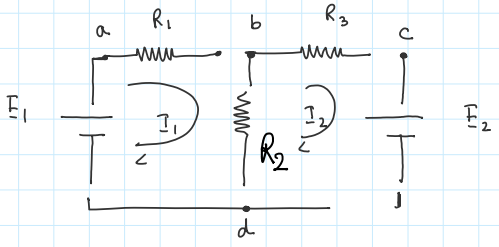
If $\underline{I}_1$ is response due to $\underline{E}_1$, $\underline{I}_2$ is the response due to $\underline{E}_2$, then in any linear circuit, the response due to the combined forcing function $(\underline{E}_1, \underline{E}_2)$ is $(\underline{I}_1, \underline{I}_2)$.

The theorem is quite general in nature. The only requirement is to have linear relationship between the cause and effect.

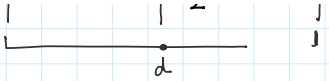
Network Analysis by Mesh Currents:

Consider the following circuit:

The following important terms are defined.



The following important terms are defined.



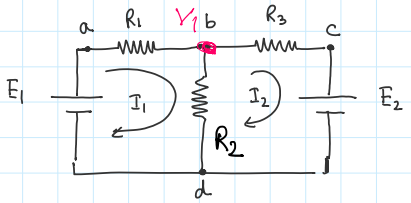
Node (n): is the point at which two or more circuit elements are joined. For example in above circuit, a, b, c, d are nodes.

For this circuit $n = 4$.

Junction (j): A junction is the point at which three or more circuit elements joined. In the circuit, b, d are junctions.

Thus $j = 2$

Branch (b): A branch is a part of network that lies between two junction points. In the above circuit, b-c-d, b-d, b-a-d are branches.



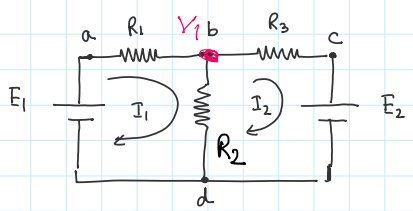
Loop: A loop is any closed path of the network. For example in the circuit a-b-d-a, b-c-d-b, a-b-c-d-a

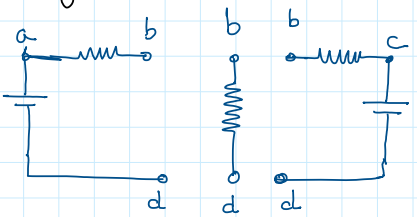
Mesh (m): A mesh is most elementary form of loop. Thus mesh can not be further divided in loops, but loops can qualify as mesh. The equation for mesh is written below.

$$m = b - (j-1) = 3 - (2-1) = 3 - 1 = 2$$

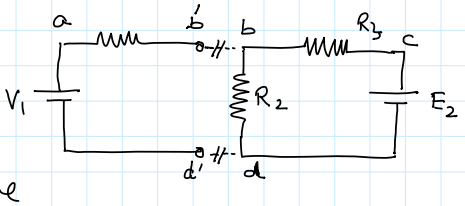
Tree: A tree is defined as the part of network composed of those branches, including all junction points, without forming a closed path.

For example consider the following circuit:

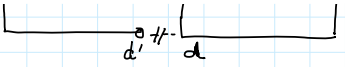




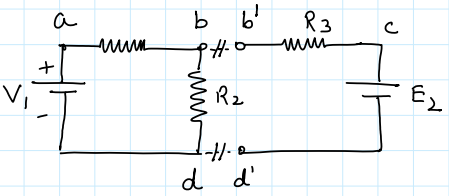
I: If branch b-a-d is drawn, then we have left closed circuit bcd



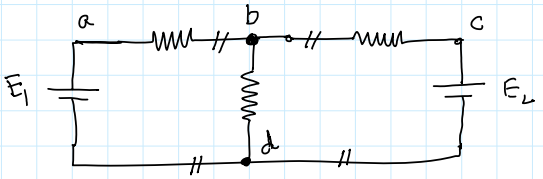
II: If branch b-c-d is drawn, we



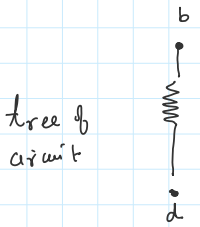
II : If branch b-c-d is drawn, we have closed path a-b-d. thus b-c-d does not form tree.



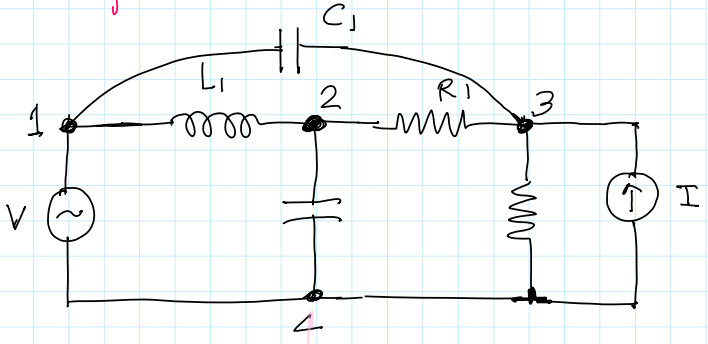
III If we draw branch b-d, then we have to cut wire from both side, to take it out like this

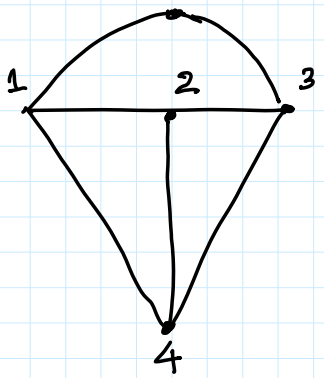


A tree refers open circuit paths containing all junctions. For example in the circuit, branch b-d is a tree.



Example 2: Consider this circuit:





Number of junction $j = 4$

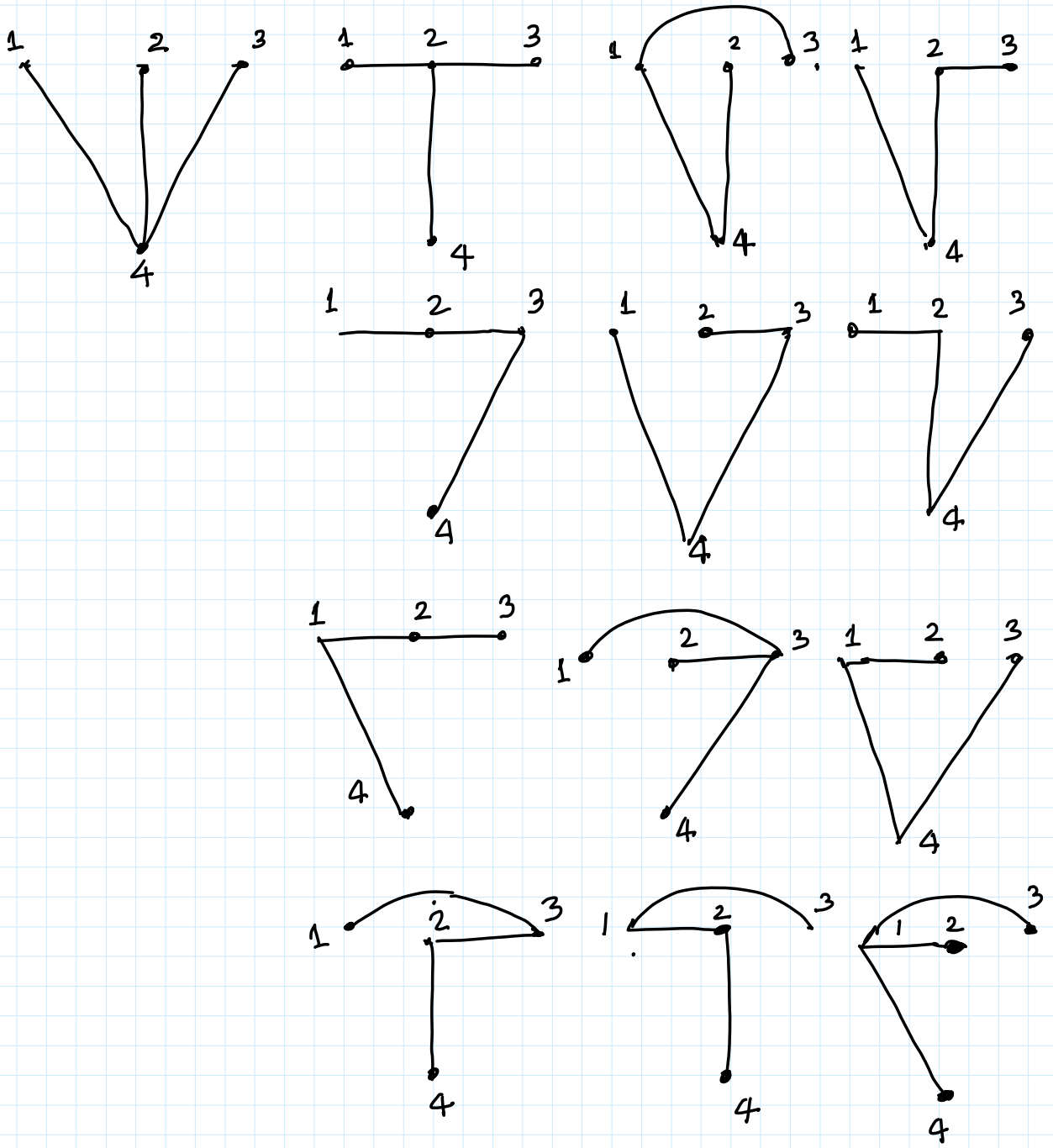
Number of branches $b = 6$

voltage source is considered as short circuit

Current source is considered as open circuit

$m = b - (j-1) = 6 - (4-1) = 3$

Various trees configurations are given:



Example 3: Consider the circuit:

Total number of junctions = 5

Total number of branches = 8

Number of meshes = $m = b - (j-1)$

$= 8 - (5-1)$

$= 4$

Number of node variables = $j - 1 = 4$

Thus both mesh and nodal analysis requires same number of variables.

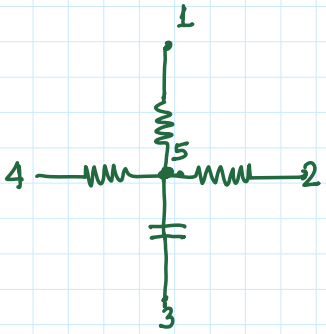
Tress and Co-tree in the network:

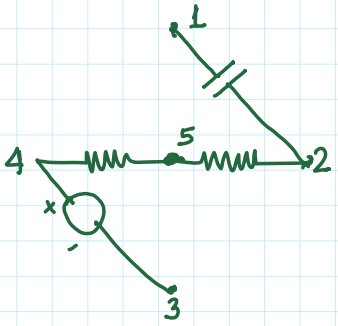
Tress and Co-tree in the network:

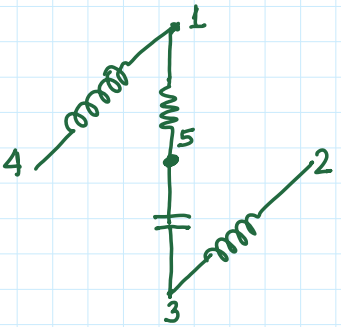
1. Remove branches:
   1-2, 2-3, 3-4, 4-1

2: 1-2 $\overset{5}{}$ 4 3

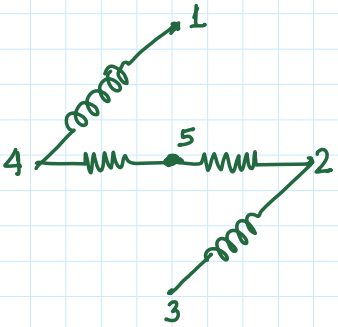
3: 2-3 $\overset{5}{}$ 1-4



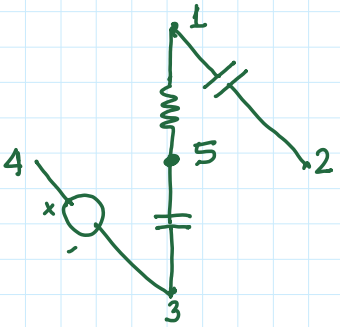


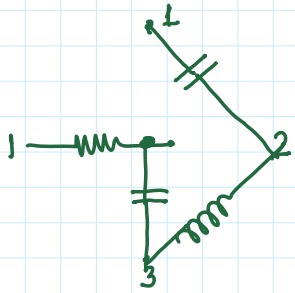


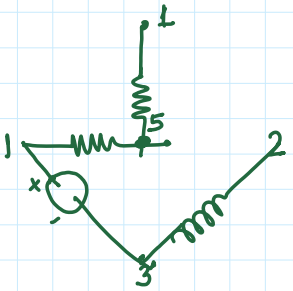
6: 1-4 $\overset{5}{}$ 2-3

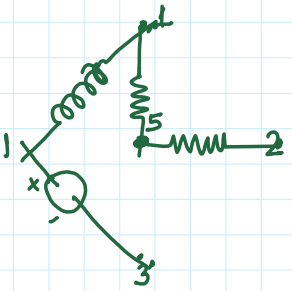


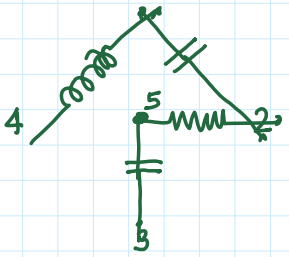
7: 2-1 $\overset{5}{}$ 3-4

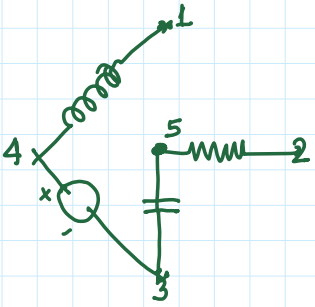


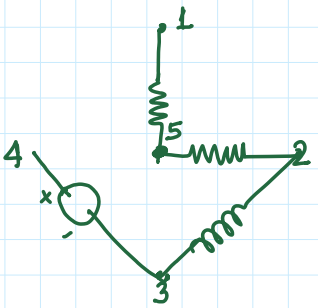


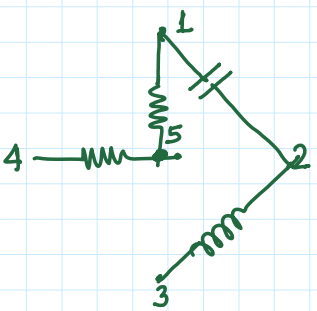


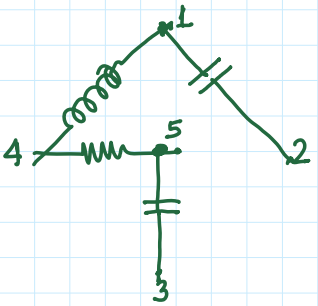


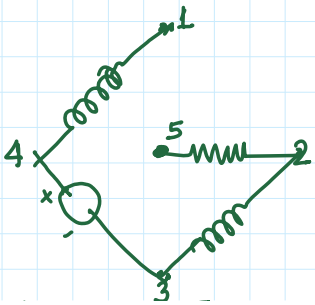


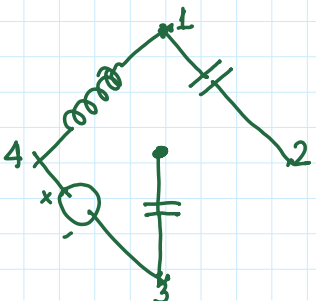


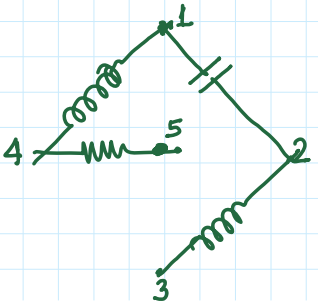


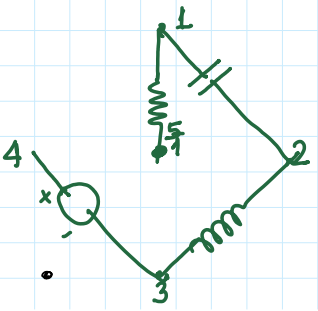


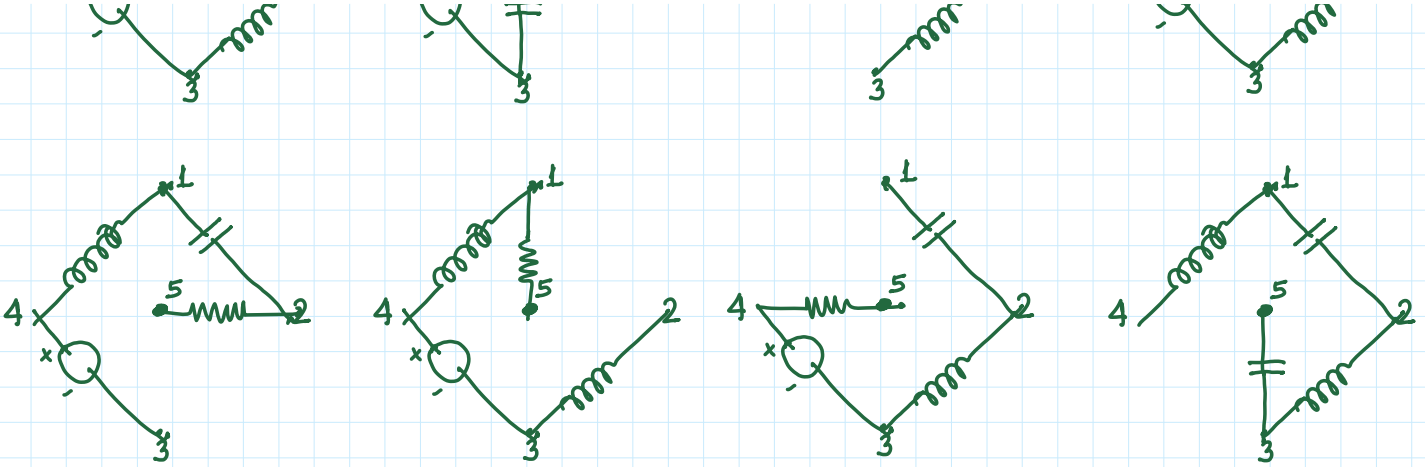











1
2
3
4
5
x